

LESSON PLAN	COURSE: MHF4U
Date: Lesson #3	Unit of Study: Chapter #2 Polynomial Equations and Inequalities
Materials / Resources: - Advanced Function 12 Study Guide - Chapter #2 Formula, Aid and Homework Sheet(s)	Lesson Focus - The Remainder Theorem - The Factor Theorem - Polynomial Equations
Prior Knowledge: - Long division with polynomial functions - Factors of polynomials (i.e. roots) and how they relate to the x intercepts	
Overall Expectations: 1. Identify and describe some key features of polynomial functions and make connections between numeric, graphical, and algebraic representations of polynomial functions	
Specific Expectations: 1.5 Make connections, through investigation using graphing technology (e.g., dynamic geometry software), between a polynomial function given in factored form [e.g., $f(x) = 2(x - 3)(x + 2)(x - 1)$] and the x-intercepts of its graph, and sketch the graph of a polynomial function given in factored form using its key features (e.g., by determining intercepts and end behaviour; by locating positive and negative regions using test values between and on either side of the x-intercepts) 1.8 Determine the equation of the family of polynomial functions with a given set of zeros and of the member of the family that passes through another given point [e.g., a family of polynomial functions of degree 3 with zeros 5, -3, and -2 is defined by the equation $f(x) = k(x - 5)(x + 3)(x + 2)$, where k is a real number, $k \neq 0$; the member of the family that passes through (-1, 24) is $f(x) = -2(x - 5)(x + 3)(x + 2)$] 3.1 Make connections, through investigation using technology (e.g., computer algebra systems), between the polynomial function $f(x)$, the divisor $x - a$, the remainder from the division, and $f(a)$ to verify the remainder theorem and the factor theorem 3.2 factor polynomial expressions in one variable, of degree no higher than four, by selecting and applying strategies (i.e., common factoring, difference of squares, trinomial factoring, factoring by grouping, remainder theorem, factor theorem) 3.3 determine, through investigation using technology (e.g., graphing calculator, computer algebra systems), the connection between the real roots of a polynomial equation and the x-intercepts of the graph of the corresponding polynomial function, and describe this connection [e.g., the real roots of the equation $x^4 - 13x^2 + 36 = 0$ are the x-intercepts of the graph of $f(x) = x^4 - 13x^2 + 36$] 3.4 solve polynomial equations in one variable, of degree no higher than four (e.g., $2x^3 = 3x^2 + 8x - 12 = 0$), by selecting and applying strategies (i.e., common factoring, difference	

of squares, trinomial factoring, factoring by grouping, remainder theorem, factor theorem), and verify solutions using technology (e.g., using computer algebra systems to determine the roots; using graphing technology to determine the x-intercepts of the graph of the corresponding polynomial function)

Learning Goals:

- Students will be able to apply the Remainder Theorem when dividing polynomials by a binomial and understands the meaning of the constant remainder
- Students will be able to apply the Factor Theorem when dividing polynomials by binomials and understands that if $f(a) = 0$, then $(x - a)$ is a factor
- Students will be able to solve polynomial equations using a variety of tactics
- Students will understand the connections between roots of polynomials and x-intercepts of polynomials and understand how they are the same values

Success Criteria:

1. By the end of this lesson I will be able to apply the Remainder Theorem to determine the remainder when a polynomial is divided by a binomial
2. By the end of this lesson I will be able to apply the Factor Theorem to factor polynomials and understand that this is a special case of the Remainder Theorem
3. By the end of this lesson I will be able to solve polynomial equations using a variety of strategies
4. By the end of this lesson I will be able to describe the connection between the real roots of a polynomial equation and the x-intercepts of the graph of the corresponding function

Assessment Strategies:

Formative: homework for chapters 2.1, 2.2 & 2.3

Summative: quiz #2, test #1

Time:

Part 1

Introduction:

Students will be given a pre-assessment quiz to determine their knowledge standings for polynomial functions, and then proceed from their answers.

Time:

Part 2

Lesson:

Chapter 2.1 The Remainder Theorem

Give a quick example of long division using real numbers and each part that is associated with the process (e.g. divisor, quotient & remainder). Explain the process of the remainder theorem using long division and synthetic division and give an example using both procedures. Compare and contrast both procedures.

Chapter 2.2 The Factor Theorem

Explain the process of the factor theorem and how it relates to the remainder theorem. Guide students to notice that the factor theorem is merely a special case of the

	remainder theorem where the divisor is a factor of the polynomial. Draw attention to the divisor as it is also a root of the polynomial. <u>Chapter 2.3 Polynomial Functions</u> Give an example of the roots of a polynomial versus its x-intercepts (e.g. a linear equation). Establish the roots of a polynomial equation make the equation equals to zero, which is the y value and hence it is an x-intercept. Solve polynomial equations using a variety of different methods such as common factoring, difference of squares, trinomial factoring, factoring by grouping, remainder theorem, factor theorem.
Time: Part 3	Wrap-up: Ask students to demonstrate their acquired knowledge by applying them to example problems and then answer any unresolved questions.
Notes, reminders & homework: <u>Reminder:</u> - Quiz #2 (to be finished before/after lesson #4) <u>Homework:</u> Chapter 2.1: # 1 – 14 Chapter 2.2: # 1 – 7, 9 - 13 Chapter 2.3: # 1 – 9	
Lesson Reflection:	

